

Lab 4: Unit-root tests and ARIMA modelling

Reference solution

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Reference Python code in `correction.py`. All numerical values below are obtained with `numpy.random.seed(42)`.

Exercise 1 – TS vs. DS by simulation

Solution

Both series exhibit an upward drift around $0.05t$, but their deviations behave very differently:

	TS: $Y_t = 0.05t + \varepsilon_t$	DS: $Z_t = Z_{t-1} + 0.05 + \eta_t$
OLS-detrended residual sd	0.882	3.112

The TS detrended residual is essentially $\mathcal{N}(0, 1)$ noise; the DS residual still wanders because OLS detrending cannot remove the unit root — it only removes a deterministic line.

Exercise 2 – ADF test

Solution

Series	ADF stat	p -value	Decision at 5%
Y_t (TS, regression <code>ct</code>)	−9.098	0.0000	reject H_0 – stationary
Z_t (RW+drift)	−0.687	0.8500	fail to reject – unit root
ΔZ_t	−13.507	0.0000	reject – stationary

The ADF test correctly identifies Y_t as stationary around a trend, Z_t as having a unit root, and one differencing as enough to remove the unit root.

Exercise 3 – KPSS test

Solution

Series	KPSS stat	p -value	Decision at 5%
Y_t (TS, <code>ct</code>)	0.092	≥ 0.10	fail to reject – stationary
Z_t (RW+drift, <code>c</code>)	1.789	≤ 0.01	reject – unit root

ADF and KPSS *agree* on both series:

- For Y_t : ADF rejects unit root *and* KPSS does not reject stationarity $\Rightarrow$ **trend-stationary** (top-left cell of the combination matrix).

- For Z_t : ADF fails to reject unit root *and* KPSS rejects stationarity $\Rightarrow$ **difference-stationary** (bottom-right cell).

Exercise 4 – Differencing the random walk

Solution

The empirical ACF tells the story:

ACF at lags 1–5					
Z_t	0.982,	0.963,	0.946,	0.929,	0.913
ΔZ_t	+0.039,	−0.069,	−0.008,	−0.000,	+0.024

$\rho_Z(h)$ decays *linearly* (the unit-root signature), while $\rho_{\Delta Z}(h)$ is essentially zero — one differencing has turned the random walk into white noise.

Re-applying the tests on ΔZ_t :

ADF: stat = −13.51, $p = 0.000$ (reject unit root), KPSS: stat = 0.101, $p \geq 0.10$ (stationary).

Both tests now agree on stationarity. The series Z_t is $I(1)$ and $\Delta Z_t \sim WN$.

Exercise 5 – SARIMA on Souvenir sales

Solution

(1)–(2) On the raw $\log Y_t$ ($T = 84$):

Test	Statistic	p -value	Decision
ADF, regression ct	−1.046	0.938	fail to reject – unit root
KPSS, regression ct	+0.117	≥ 0.10	fail to reject – stationary

The two tests are *ambiguous* but consistent with the near-unit-root behaviour expected from a strongly trending, seasonal economic series. We proceed with one regular and one seasonal differencing.

(3) SARIMA(0, 1, 1)(0, 1, 1)₁₂ fitted by maximum likelihood (**statsmodels**):

$$\hat{\theta}_1 = -0.4991, \quad \hat{\Theta}_1 = -0.3245, \quad \hat{\sigma}^2 = 0.0247, \quad \text{AIC} = -43.08, \quad \text{BIC} = -36.95.$$

(4) Ljung–Box on the residuals at lag 24:

$$Q_{LB}(24) = 16.04, \quad p\text{-value} = 0.887.$$

We **do not reject** the white-noise hypothesis: the residuals have no remaining autocorrelation, so the SARIMA captures the structure of the series.

(5) The 12-month forecast (figure **lab4_ex5_forecast.png**) extrapolates the upward trend and the December peak; the 95% confidence interval widens with the forecast horizon, as expected for an integrated process.